

Bioassay Suite — GUI Refresh · Implementation Guide v2

Read this first. v1 was implemented partially and the result drifted from the mockup. This v2 is **prescriptive** — it gives you full, ready-to-paste files instead of guidelines. Do not stop halfway: every section below is required for the design to read as finished.

Ground rules (unchanged): 1. **No business logic, server reactives, analysis pipeline, or Rmd report-token CSS changes.** Do not touch `drc` fitting, outlier detection, tissue normalization, multi-wavelength concordance, auto-save, file-import logic, output folder naming, or `runApp()` entry points. 2. **All four plate matrices stay** (`matrix_type`, `matrix_id`, `matrix_dilution`, `matrix_replicate`). They keep all observers, change-handlers, and `shared$matrix_*` reactives. Switching layers is **visibility only** — no rebuild, no data loss. 3. **Bilingual translation engine stays.** Only **add** keys. 4. If anything in this guide conflicts with existing logic, **stop and ask** — do not improvise.

The visual reference is `Bioassay Suite – GUI refresh.html` (open in a browser; pan/zoom canvas; click any artboard to focus fullscreen). Token CSS lives in `tokens.css`.

What went wrong in v1 (so you know what to avoid)

Looking at the v1 build screenshots:

SYMPTOM	ROOT CAUSE	FIX IN V2
Wizard tab text invisible (white on white)	<code>.nav-pills > li > a</code> color was overridden but the <code></code> inside Bootstrap's pill markup wasn't	§ 2.4 — full stepper CSS with <code>color</code> + <code>background</code> set on every state
No progress connector between steps	Connectors weren't implemented at all	§ 2.4 — <code>::after</code> pseudo-element on every <code></code> except <code>:last-child</code>
"Bioassay Suite v0.9.0" rendering as broken text	The new top-bar HTML wasn't built; only the wordmark was changed in place	§ 2.2 — full top-bar <code>div</code> replaces the old block; copy verbatim
Quick Start became 3 plain Bootstrap buttons (no tile design)	The 6→3 collapse happened server-side but UI markup stayed as <code>actionButtons</code>	§ 3.1 — replace with <code>actionLink(class = "preset-tile", HTML(...))</code>
Assay/Standards labels not paired with inputs	The "two-column row" pattern wasn't built	§ 3.2 — explicit <code>fluidRow + column(4) + column(8)</code> skeleton
Plate Layout uses radio buttons, not the layer switcher	The CSS to restyle radios as a vertical tablist wasn't added	§ 4.2 — full CSS block, plus markup change to wrap inputs in a labeled <code>div.bs-layer-switcher</code>
Tab 3 has no Import Summary card; heatmap is below the data table	The two-column layout wasn't built — content is stacked single-column	§ 5.1 — explicit <code>fluidRow</code> skeleton with summary card on the right
Heatmap palette still RdBu / yellow-blue	<code>plot_functions.R</code> palette swap was skipped	§ 5.1 — exact line to add to the heatmap call

Bottom line for the implementer: if you read a section and think "I'll do the CSS but skip the markup change because Shiny makes it awkward" — **don't**. The markup changes ARE the redesign. If a Shiny pattern blocks you, ask before improvising.

§ 1 • Replace `www/style.css` entirely

Delete the current file. Paste the block below. **Do not selectively port rules.** The variables and selectors here are referenced by every section that follows.

```

/* =====
Bioassay Suite v0.9.0 – Design Tokens & Components
Drop-in replacement for www/style.css
===== */

:root {
  /* – Type – */
  --ff-sans: "IBM Plex Sans", system-ui, -apple-system, "Segoe UI", Roboto, sans-serif;
  --ff-mono: "IBM Plex Mono", ui-monospace, "SF Mono", Menlo, monospace;
  --ff-serif: "IBM Plex Serif", Georgia, serif;

  /* – Neutrals – */
  --c-bg: #f7f8f9;
  --c-surface: #ffffff;
  --c-surface-2: #f1f3f5;
  --c-line: #e3e6ea;
  --c-line-2: #eef0f3;
  --c-ink: #0f1722;
  --c-ink-2: #2c3441;
  --c-ink-3: #5b6573;
  --c-ink-4: #8a93a1;

  /* – Brand accent (deep navy) – */
  --c-accent: #1f3a5f;
  --c-accent-2: #16294a;
  --c-accent-soft: #e8edf4;

  /* – Semantic (Okabe-Ito, colorblind-safe) – */
  --c-pass: #009e73; --c-pass-soft: #e6f5ee;
  --c-warn: #e69f00; --c-warn-soft: #fbbf0d9;
  --c-fail: #d55e00; --c-fail-soft: #fbbf0d9;
  --c-info: #0072b2; --c-info-soft: #dfecf5;

  /* – Geometry – */
  --r-1: 3px; --r-2: 6px; --r-3: 10px;
  --sh-1: 0 1px 2px rgba(15,23,34,0.06), 0 0 1px rgba(15,23,34,0.04);

  /* – Spacing scale – */
  --s-1: 4px; --s-2: 8px; --s-3: 12px; --s-4: 16px;
  --s-5: 24px; --s-6: 32px; --s-7: 48px;
}

/* – Reset & base – */
html, body {
  background: var(--c-bg);
  color: var(--c-ink);
  font-family: var(--ff-sans);
  font-size: 13px;
  line-height: 1.55;
  -webkit-font-smoothing: antialiased;
}

a { color: var(--c-accent); text-decoration: none; }
a:hover { text-decoration: underline; }

code, kbd, samp, pre { font-family: var(--ff-mono); }

/* sr-only kept */

```

```

.sr-only { position: absolute; width: 1px; height: 1px; padding: 0; margin: -1px;
  overflow: hidden; clip: rect(0,0,0,0); border: 0; }

/* =====
§ 2.2 – Top bar
===== */
.bs-topbar {
  display: flex; align-items: center; justify-content: space-between;
  padding: 14px 28px;
  background: var(--c-surface);
  border-bottom: 1px solid var(--c-line);
  margin: 0 -15px 0; /* cancel Shiny container padding */
}
.bs-topbar .brand { display: flex; align-items: center; gap: 12px; }
.bs-topbar .brand .mono-tile {
  width: 26px; height: 26px; border-radius: var(--r-1);
  background: var(--c-accent); color: #fff;
  display: grid; place-items: center;
  font-family: var(--ff-mono); font-weight: 600; font-size: 13px;
}
.bs-topbar .brand .wordmark {
  font-size: 15px; font-weight: 600; color: var(--c-ink);
  letter-spacing: -0.005em;
}
.bs-topbar .brand .version {
  font-family: var(--ff-mono); font-size: 11px; color: var(--c-ink-4);
  background: var(--c-surface-2); padding: 2px 7px; border-radius: 999px;
}
.bs-topbar .actions { display: flex; align-items: center; gap: 14px; }
.bs-topbar .actions .quiet-btn {
  font-size: 12px; color: var(--c-ink-3); padding: 6px 10px;
  background: transparent; border: 0; cursor: pointer;
}
.bs-topbar .actions .quiet-btn:hover { color: var(--c-ink); background: var(--c-surface-2); border-
radius: var(--r-1); }
.bs-topbar .actions .divider { width: 1px; height: 18px; background: var(--c-line); }
.bs-topbar .actions select {
  font-family: var(--ff-sans); font-size: 12px; color: var(--c-ink-2);
  background: var(--c-surface); border: 1px solid var(--c-line);
  border-radius: var(--r-1); padding: 5px 8px;
}

/* =====
§ 2.4 – Numbered stepper (replaces nav-pills)
This restyles the existing tabsetPanel(type="pills") markup.
===== */
.nav-pills {
  display: flex !important;
  align-items: center;
  gap: 0;
  padding: 18px 28px 22px;
  margin: 0 -15px 0;
  background: var(--c-surface);
  border-bottom: 1px solid var(--c-line);
  list-style: none;
}
.nav-pills > li { flex: 1; position: relative; margin: 0; }
.nav-pills > li:not(:last-child)::after {
  content: ""; position: absolute;

```

```

top: 50%; right: 0; left: calc(100% - 0px);
height: 1px; background: var(--c-line);
width: 100%; max-width: 48px;      /* short connector */
pointer-events: none;
}
.nav-pills > li > a {
  display: flex; align-items: center; gap: 10px;
  padding: 8px 14px;
  background: transparent !important;
  border: 1px solid var(--c-line) !important;
  border-radius: 999px !important;
  font-size: 13px !important;
  font-weight: 500 !important;
  color: var(--c-ink-3) !important;      /* fixes the white-on-white bug */
  transition: background 120ms, color 120ms, border 120ms, box-shadow 120ms;
}
.nav-pills > li > a::before {
  content: attr(data-step);
  font-family: var(--ff-mono);
  font-weight: 500; font-size: 12px;
  width: 22px; height: 22px;
  display: grid; place-items: center;
  border-radius: 999px;
  background: var(--c-surface-2);
  color: var(--c-ink-3);
  flex-shrink: 0;
}
.nav-pills > li > a:hover {
  color: var(--c-ink) !important;
  border-color: var(--c-ink-4) !important;
  background: var(--c-surface-2) !important;
}
.nav-pills > li.is-done > a::before {
  background: var(--c-accent);
  color: #fff;
  content: "✓";
  font-family: var(--ff-sans);
  font-weight: 600;
}
.nav-pills > li.is-done > a {
  color: var(--c-ink-2) !important;
  border-color: var(--c-line) !important;
}
.nav-pills > li.active > a,
.nav-pills > li.active > a:hover,
.nav-pills > li.active > a:focus {
  background: var(--c-accent-soft) !important;
  border-color: var(--c-accent) !important;
  color: var(--c-accent) !important;
  box-shadow: 0 0 0 3px var(--c-accent-soft);
}
.nav-pills > li.active > a::before {
  background: var(--c-accent);
  color: #fff;
}
}

/* =====
§ 2.5 – Cards & sections
===== */

```

```

.bs-card {
  background: var(--c-surface);
  border: 1px solid var(--c-line);
  border-radius: var(--r-2);
  box-shadow: var(--sh-1);
  padding: 20px 22px;
}
.bs-card + .bs-card { margin-top: var(--s-5); }
.bs-card-title {
  font-size: 14px; font-weight: 600; color: var(--c-ink);
  margin: 0 0 4px;
}
.bs-card-sub {
  font-size: 12px; color: var(--c-ink-3);
  margin: 0 0 16px;
}

.bs-section-row {
  display: grid;
  grid-template-columns: minmax(220px, 320px) 1fr;
  gap: var(--s-6);
  align-items: start;
  padding: 22px 0;
  border-top: 1px solid var(--c-line);
}
.bs-section-row:first-of-type { border-top: 0; padding-top: 8px; }
.bs-section-row .label-col h4 {
  font-size: 14px; font-weight: 600; color: var(--c-ink);
  margin: 0 0 4px;
  border: 0; padding: 0; /* override Bootstrap */
}
.bs-section-row .label-col p {
  font-size: 12px; color: var(--c-ink-3); margin: 0;
}

/* =====
   § 3.1 – Quick Start preset tiles
   ===== */
.bs-quickstart {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: var(--s-4);
  margin-top: var(--s-3);
}
.preset-tile {
  display: flex; align-items: stretch;
  background: var(--c-surface);
  border: 1px solid var(--c-line);
  border-radius: var(--r-2);
  overflow: hidden;
  text-align: left;
  cursor: pointer;
  transition: border-color 120ms, box-shadow 120ms, transform 120ms;
  text-decoration: none !important;
  color: var(--c-ink) !important;
}
.preset-tile:hover {
  border-color: var(--c-accent);
  box-shadow: 0 0 0 3px var(--c-accent-soft);
}

```

```

}
.preset-tile .rail { width: 4px; flex-shrink: 0; }
.preset-tile .rail-blue { background: var(--c-info); }
.preset-tile .rail-green { background: var(--c-pass); }
.preset-tile .rail-purple { background: #cc79a7; }
.preset-tile .body { padding: 14px 16px; flex: 1; }
.preset-tile .title { font-size: 13px; font-weight: 600; color: var(--c-ink); margin-bottom: 2px; }
.preset-tile .sub { font-size: 11.5px; color: var(--c-ink-3); }
.preset-tile .meta {
  margin-top: 8px;
  font-family: var(--ff-mono); font-size: 10.5px;
  color: var(--c-ink-4); text-transform: uppercase; letter-spacing: 0.06em;
}

/* =====
§ 4.2 – Plate Layout layer switcher
Restyles a radioButtons() group as a vertical tablist.
===== */
.bs-layout-shell {
  display: grid;
  grid-template-columns: 240px 1fr;
  gap: var(--s-5);
}
.bs-layer-switcher {
  background: var(--c-surface);
  border: 1px solid var(--c-line);
  border-radius: var(--r-2);
  padding: 12px;
  align-self: start;
}
.bs-layer-switcher h5 {
  font-family: var(--ff-mono); font-size: 10.5px; font-weight: 600;
  text-transform: uppercase; letter-spacing: 0.08em;
  color: var(--c-ink-4);
  margin: 4px 8px 8px;
  border: 0; padding: 0;
}
.bs-layer-switcher .shiny-options-group { display: flex; flex-direction: column; gap: 2px; }
.bs-layer-switcher .radio { margin: 0 !important; padding: 0 !important; }
.bs-layer-switcher .radio label {
  display: flex; align-items: center; gap: 10px;
  padding: 8px 10px;
  border-radius: var(--r-1);
  font-size: 13px; font-weight: 500;
  color: var(--c-ink-2);
  cursor: pointer;
  position: relative;
  width: 100%;
}
.bs-layer-switcher .radio label::before {
  content: attr(data-step);
  font-family: var(--ff-mono); font-size: 11px; font-weight: 500;
  color: var(--c-ink-4);
  width: 22px; flex-shrink: 0;
}
.bs-layer-switcher .radio label:hover { background: var(--c-surface-2); }
.bs-layer-switcher .radio input[type="radio"] {
  position: absolute; opacity: 0; pointer-events: none;
}

```

```

.bs-layer-switcher .radio input[type="radio"]:checked + span {
  /* the span wraps the label text in Shiny radioButtons markup */
  color: var(--c-accent);
}
.bs-layer-switcher .radio:has(input:checked) label {
  background: var(--c-accent-soft);
  color: var(--c-accent);
}
.bs-layer-switcher .radio:has(input:checked) label::before {
  color: var(--c-accent);
  font-weight: 600;
}

.bs-layer-bulk {
  margin-top: 14px; padding-top: 14px;
  border-top: 1px solid var(--c-line);
}
.bs-layer-bulk h6 {
  font-family: var(--ff-mono); font-size: 10.5px; font-weight: 600;
  text-transform: uppercase; letter-spacing: 0.08em;
  color: var(--c-ink-4);
  margin: 0 8px 8px;
}

.bs-matrix-viewport { min-width: 0; overflow-x: auto; }

.bs-legend {
  display: flex; flex-wrap: wrap; gap: 14px;
  margin-top: var(--s-3); padding: 10px 12px;
  background: var(--c-surface-2);
  border-radius: var(--r-1);
  font-size: 12px;
}
.bs-legend .chip { display: inline-flex; align-items: center; gap: 6px; color: var(--c-ink-2); }
.bs-legend .dot { width: 10px; height: 10px; border-radius: 2px; border: 1px solid rgba(0,0,0,0.08); }

/* =====
   § 5 – Forms & inputs
   ===== */
.form-group label,
.control-label {
  font-size: 12px; font-weight: 500; color: var(--c-ink-2);
  margin-bottom: 4px;
}
.form-control, .selectize-input, input[type="text"], input[type="number"], select, textarea {
  font-family: var(--ff-mono);
  font-size: 12px;
  height: 32px;
  border: 1px solid var(--c-line) !important;
  border-radius: var(--r-1) !important;
  background: var(--c-surface) !important;
  color: var(--c-ink) !important;
  box-shadow: none !important;
  padding: 4px 10px;
}
.form-control:focus, .selectize-input:focus {
  border-color: var(--c-accent) !important;
  box-shadow: 0 0 0 3px var(--c-accent-soft) !important;
}

```



```

textarea, .form-control[rows] { height: auto; }

/* Numeric/standard inputs in 4-column grid */
.bs-std-grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: var(--s-3) var(--s-4);
}

/* =====
   § 5 – Buttons
   ===== */
.btn, .btn-default, .btn-primary, .btn-success {
  font-family: var(--ff-sans);
  font-size: 13px; font-weight: 500;
  height: 34px;
  padding: 0 14px;
  border-radius: var(--r-1) !important;
  box-shadow: none !important;
  transition: background 120ms, color 120ms, border 120ms;
}
.btn-default {
  background: var(--c-surface) !important;
  border: 1px solid var(--c-line) !important;
  color: var(--c-ink-2) !important;
}
.btn-default:hover {
  background: var(--c-surface-2) !important;
  border-color: var(--c-ink-4) !important;
  color: var(--c-ink) !important;
}
.btn-primary {
  background: var(--c-accent) !important;
  border: 1px solid var(--c-accent) !important;
  color: #fff !important;
}
.btn-primary:hover {
  background: var(--c-accent-2) !important;
  border-color: var(--c-accent-2) !important;
}
.btn-success {
  background: var(--c-pass) !important;
  border: 1px solid var(--c-pass) !important;
  color: #fff !important;
}
.btn-block { width: 100%; display: block; }

/* =====
   § 5 – Pills, chips, status
   ===== */
.bs-pill {
  display: inline-flex; align-items: center; gap: 5px;
  padding: 2px 8px; border-radius: 999px;
  font-size: 11px; font-weight: 500;
  font-family: var(--ff-sans);
}
.bs-pill.is-pass { background: var(--c-pass-soft); color: var(--c-pass); }
.bs-pill.is-warn { background: var(--c-warn-soft); color: var(--c-warn); }
.bs-pill.is-fail { background: var(--c-fail-soft); color: var(--c-fail); }

```

```

.bs-pill.is-info { background: var(--c-info-soft); color: var(--c-info); }

/* =====
   § 5 – Well / panel mapping
   ===== */
.well, .panel, .panel-default {
  background: var(--c-surface) !important;
  border: 1px solid var(--c-line) !important;
  border-radius: var(--r-2) !important;
  box-shadow: var(--sh-1) !important;
}

/* =====
   § 5 – Headings (override the old blue-bordered h4/h5)
   ===== */
h1, h2, h3, h4, h5 {
  color: var(--c-ink);
  border: 0;
  padding: 0;
  margin: 0 0 8px;
}
h2 { font-size: 18px; font-weight: 600; }
h3 { font-size: 16px; font-weight: 600; }
h4 { font-size: 14px; font-weight: 600; }
h5 { font-size: 12.5px; font-weight: 600; color: var(--c-ink-2); }

.eyebrow {
  font-family: var(--ff-mono); font-size: 10.5px; font-weight: 500;
  text-transform: uppercase; letter-spacing: 0.1em;
  color: var(--c-ink-4); margin-bottom: 6px;
}

/* =====
   § 5 – Handsontable
   ===== */
.handsontable {
  font-family: var(--ff-mono) !important;
  font-size: 11px !important;
}
.handsontable th {
  background: var(--c-surface-2) !important;
  color: var(--c-ink-3) !important;
  font-weight: 500 !important;
  font-family: var(--ff-mono) !important;
  font-size: 10.5px !important;
  text-transform: uppercase;
  letter-spacing: 0.04em;
  border-color: var(--c-line) !important;
}
.handsontable td {
  border-color: var(--c-line-2) !important;
  color: var(--c-ink-2) !important;
}

/* – Sample-Type cell colors (Okabe-Ito tinted, CB-safe) – */
.handsontable td.cell-standard { background: rgba(0,158,115,0.10) !important; color: #006d51 !important; }
.handsontable td.cell-sample { background: rgba(0,114,178,0.10) !important; color: #00497a !important; }

```

```

.handsonstable td.cell-blank    { background: var(--c-surface-2) !important; color: var(--c-ink-3) !important; }
.handsonstable td.cell-nsb     { background: rgba(204,121,167,0.12) !important; color: #8a4a6e !important; }
.handsonstable td.cell-b0      { background: rgba(230,159,0,0.12) !important; color: #8a5e00 !important; }

/* =====
   § 5 – Pre-flight check panel
   ===== */
.bs-preflight { display: flex; flex-direction: column; gap: 8px; }
.bs-preflight .item {
  display: flex; align-items: flex-start; gap: 10px;
  padding: 10px 12px;
  background: var(--c-surface);
  border: 1px solid var(--c-line);
  border-radius: var(--r-1);
}
.bs-preflight .glyph {
  width: 22px; height: 22px; border-radius: 999px;
  display: grid; place-items: center;
  font-size: 12px; font-weight: 600; flex-shrink: 0;
}
.bs-preflight .item.is-pass .glyph { background: var(--c-pass-soft); color: var(--c-pass); }
.bs-preflight .item.is-warn .glyph { background: var(--c-warn-soft); color: var(--c-warn); }
.bs-preflight .item.is-fail .glyph { background: var(--c-fail-soft); color: var(--c-fail); }
.bs-preflight .desc { font-size: 12.5px; color: var(--c-ink-2); }
.bs-preflight .meta {
  font-family: var(--ff-mono); font-size: 11px; color: var(--c-ink-4);
  margin-top: 2px;
}

/* =====
   § 5 – Format / language tiles
   ===== */
.bs-tile-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(140px, 1fr));
  gap: var(--s-3);
}
.bs-tile-check {
  position: relative;
  border: 1px solid var(--c-line);
  border-radius: var(--r-2);
  padding: 12px 14px;
  background: var(--c-surface);
  cursor: pointer;
}
.bs-tile-check input { position: absolute; opacity: 0; pointer-events: none; }
.bs-tile-check:has(input:checked) {
  border-color: var(--c-accent);
  background: var(--c-accent-soft);
}
.bs-tile-check .name { font-size: 13px; font-weight: 600; color: var(--c-ink); }
.bs-tile-check .ext  { font-family: var(--ff-mono); font-size: 11px; color: var(--c-ink-4); }

/* =====
   § 6 – Report KPI strip (in Rmd output too)
   ===== */

```

```

.bs-kpis {
  display: grid;
  grid-template-columns: repeat(5, 1fr);
  gap: var(--s-3);
  margin: 18px 0 24px;
}
.bs-kpi {
  background: var(--c-surface);
  border: 1px solid var(--c-line);
  border-radius: var(--r-2);
  padding: 14px 16px;
}
.bs-kpi .label {
  font-family: var(--ff-mono); font-size: 10.5px;
  text-transform: uppercase; letter-spacing: 0.08em;
  color: var(--c-ink-4); margin-bottom: 6px;
}
.bs-kpi .value {
  font-family: var(--ff-mono); font-size: 22px; font-weight: 500;
  color: var(--c-ink); line-height: 1.1;
  font-variant-numeric: tabular-nums;
}
.bs-kpi .unit {
  font-family: var(--ff-mono); font-size: 11px; color: var(--c-ink-3);
  margin-top: 2px;
}
.bs-kpi.status .value { display: flex; align-items: center; gap: 8px; font-size: 16px; }
.bs-kpi .status-dot { width: 10px; height: 10px; border-radius: 999px; }
.bs-kpi.is-pass .status-dot { background: var(--c-pass); }
.bs-kpi.is-warn .status-dot { background: var(--c-warn); }
.bs-kpi.is-fail .status-dot { background: var(--c-fail); }

/* =====
   § 6 – Report executive summary, tables
   ===== */
.report-title { padding: 8px 0 24px; }
.report-title h1 {
  font-family: var(--ff-serif); font-weight: 500;
  font-size: 32px; line-height: 1.15; letter-spacing: -0.01em;
  margin: 4px 0;
}
.report-title .meta {
  font-family: var(--ff-mono); font-size: 11px; color: var(--c-ink-4);
}

table.bs-report {
  width: 100%; border-collapse: collapse; font-size: 12px;
}
table.bs-report th {
  font-family: var(--ff-mono); font-size: 10.5px; font-weight: 600;
  text-transform: uppercase; letter-spacing: 0.06em;
  color: var(--c-ink-3); background: var(--c-surface-2);
  text-align: left; padding: 8px 10px;
  border-bottom: 1px solid var(--c-line);
}
table.bs-report td {
  font-family: var(--ff-mono); font-variant-numeric: tabular-nums;
  padding: 8px 10px; border-bottom: 1px solid var(--c-line-2);
  color: var(--c-ink-2);
}

```

```

}
table.bs-report td.num { text-align: right; }

/* =====
   Responsive
   ===== */
@media (max-width: 1100px) {
  .bs-quickstart { grid-template-columns: 1fr; }
  .bs-layout-shell { grid-template-columns: 1fr; }
  .bs-section-row { grid-template-columns: 1fr; gap: var(--s-3); }
  .bs-kpis { grid-template-columns: repeat(2, 1fr); }
}
@media (max-width: 700px) {
  .nav-pills { flex-wrap: wrap; gap: 6px; padding: 12px; }
  .nav-pills > li { flex: 1 1 calc(50% - 6px); }
  .nav-pills > li:not(:last-child)::after { display: none; }
  .bs-std-grid { grid-template-columns: repeat(2, 1fr); }
}

```

§ 2 · App shell — top bar & numbered stepper

2.1 Add fonts (once)

In `app.R` `tags$head` — before the `style.css` link:

```

tags$head(
  tags$link(rel = "preconnect", href = "https://fonts.googleapis.com"),
  tags$link(rel = "preconnect", href = "https://fonts.gstatic.com", crossorigin = NA),
  tags$link(rel = "stylesheet",
    href = "https://fonts.googleapis.com/css2?
family=IBM+Plex+Mono:wght@400;500;600&family=IBM+Plex+Sans:wght@400;500;600&family=IBM+Plex+Serif:wght@
400;500;600&display=swap"),
  tags$link(rel = "stylesheet", type = "text/css", href = "style.css")
)

```

2.2 Top bar — replace the broken "Bioassay Suite v0.9.0" block

Find the current block in `app.R` that produces `uiOutput("app_title_ui")` followed by the row with `start_tour`, docs link, and language `<select>`. **Replace it with:**

```

div(class = "bs-topbar",
  div(class = "brand",
    div(class = "mono-tile", "B"),
    span(class = "wordmark", "Bioassay Suite"),
    span(class = "version", "v0.9.0")
  ),
  div(class = "actions",
    actionButton("start_tour", i18n_t("Guided tour"),
      class = "quiet-btn", icon = NULL),
    tags$a(href = "docs/index.html", target = "_blank",
      class = "quiet-btn", i18n_t("Docs")),
    div(class = "divider"),
    # keep the existing language select; the CSS in § 5 styles it
    selectInput("language", NULL,
      choices = c("English" = "en", "Español" = "es"),
      selected = isolate(input$language %||% "en"),
      width = "120px")
  )
)

```

Verify after build: the wordmark renders as a single line `[B] Bioassay Suite v0.9.0` — no broken stacking. If you see "Bioassay Suite v0.9.0" wrapping, check that the surrounding container is not a vertical `column()` — the topbar must be a direct child of `fluidPage()` and use `class = "bs-topbar"`.

2.3 Add a `data-step` attribute to each tab pill

The stepper CSS reads the step number from `data-step`. Bootstrap won't write that itself; inject it once on app start with a small JS snippet. Add to `app.R` (UI):

```

tags$script(HTML("
  $(function() {
    function stamp() {
      $('.nav-pills > li').each(function(i) {
        var n = (i + 1).toString().padStart(2, '0');
        $(this).children('a').attr('data-step', n);
      });
    }
    stamp();
    $(document).on('shown.bs.tab', stamp);
  });
"))

```

2.4 Mark done/current states

Bootstrap already adds `.active` to the current pill. You also need `.is-done` on completed pills so the check icon appears. Add **one** observer in your existing `server.R` (or wherever the `wizard_tabs` tab change is handled — do not touch the underlying logic):

```
# Stepper progress – purely cosmetic
observe({
  current <- input$wizard_tabs
  steps   <- c("config", "layout", "upload", "analysis", "report") # use your real tab values
  idx     <- match(current, steps)
  if (is.na(idx)) return()
  done_js <- sprintf("
    $('.nav-pills > li').removeClass('is-done');
    $('.nav-pills > li').slice(0, %d).addClass('is-done');
  ", idx - 1)
  shinyjs::runjs(done_js)
})
```

Requires `shinyjs::useShinyjs()` in the UI head — if not already present, add it.

§ 3 • Tab 1 — Configuration

3.1 Quick Start as 3 preset tiles (not buttons)

Find the `quickstart_section` block. **Replace** the 6-button (or the v1 3-button) markup with:

```

div(id = "quickstart_section", class = "bs-card",
  h3(i18n_t("Quick Start")),
  p(class = "bs-card-sub",
    i18n_t("Choose a preset to auto-configure the assay type, plate layout, and standard
concentrations.")),

  div(class = "bs-quickstart",

    actionLink("qs_rba_stx", class = "preset-tile", style = "padding:0;",
      div(class = "rail rail-blue"),
      div(class = "body",
        div(class = "title", "RBA · Saxitoxin"),
        div(class = "sub", i18n_t("Receptor binding assay · 8 standards · triplicate")),
        div(class = "meta", i18n_t("Demo data included")))
      )
    ),

    actionLink("qs_elisa_cortisol", class = "preset-tile", style = "padding:0;",
      div(class = "rail rail-green"),
      div(class = "body",
        div(class = "title", "ELISA · Cortisol"),
        div(class = "sub", i18n_t("Competitive ELISA · 7 standards · duplicate")),
        div(class = "meta", i18n_t("Demo data included")))
      )
    ),

    actionLink("qs_elisa_custom", class = "preset-tile", style = "padding:0;",
      div(class = "rail rail-purple"),
      div(class = "body",
        div(class = "title", "ELISA · Custom"),
        div(class = "sub", i18n_t("Blank ELISA template · configure from scratch")),
        div(class = "meta", i18n_t("No demo data")))
      )
    )
  ),

  tags$details(style = "margin-top:14px;",
    tags$summary(i18n_t("More options – configure manually without demo data")),
    div(style = "padding-top:10px; display:flex; gap:8px;",
      actionButton("qs_rba_stx_manual", i18n_t("RBA Saxitoxin – manual"), class = "btn-
default"),
      actionButton("qs_elisa_cortisol_manual", i18n_t("ELISA Cortisol – manual"), class = "btn-
default")
    )
  )
)

```

Server-side: keep all six original observers wired. Don't delete `qs_rba_stx_manual` / `qs_elisa_cortisol_manual` — they're now in the `<details>` disclosure.

3.2 Assay & Standards — paired label + input rows

The current screenshot shows section headings stacked **above** the inputs, full-width — that is wrong. Use the **two-column row** pattern. Find the assay/standards block. **Replace** with this skeleton (preserve every input ID exactly):


```

div(class = "bs-card",

  # ---- Row: Assay type ----
  div(class = "bs-section-row",
    div(class = "label-col",
      h4(i18n_t("Assay type")),
      p(i18n_t("Receptor binding or competitive ELISA – determines which curve model and toxin
standards apply."))
    ),
    div(class = "input-col",
      selectInput("assay_type", NULL,
        choices = c("Receptor Binding Assay (RBA)" = "rba",
          "ELISA" = "elisa"),
        selected = "rba"),
      conditionalPanel("input.assay_type == 'rba'",
        selectInput("toxin_standard", i18n_t("Toxin standard"),
          choices = c("Saxitoxin" = "stx", "PSP analogues" = "psp")),
        div(class = "eyebrow", i18n_t("Molecular weight (g/mol): 299.29"))
      )
    )
  ),

  # ---- Row: Standard concentrations ----
  div(class = "bs-section-row",
    div(class = "label-col",
      h4(i18n_t("Standard concentrations")),
      p(i18n_t("Specify the number of standards, then enter each concentration in mol/L using
scientific notation (e.g. 1e-6, 3e-8).")),
      div(class = "eyebrow", style = "margin-top:8px;",
        i18n_t("Tip: paste concentrations from a spreadsheet column."))
    ),
    div(class = "input-col",
      selectInput("n_standards", i18n_t("Number of standards"),
        choices = as.character(4:12), selected = "8"),
      div(class = "bs-std-grid",
        # render 8 (or n_standards) numericInputs in a 4-col grid
        # KEEP EXISTING LOGIC – wrap in this div only
        uiOutput("standards_inputs_ui")
      )
    )
  )
)

```

Do not keep the existing green `wellPanel(class = "bg-green")` info box. The eyebrow + helper text in the label column carries the same information.

§ 4 · Tab 2 — Plate Layout (the layer switcher must look like the mockup)

4.1 Layout skeleton

Find the current 2×2 grid of matrices (or the v1 radio + single-matrix layout). **Replace** with:

```

div(class = "bs-layout-shell",

# -- LEFT: layer switcher + bulk action --
div(class = "bs-layer-switcher",
  h5(i18n_t("Layers")),
  radioButtons("active_layer", NULL,
    choiceNames = list(
      tagList(span(`data-step` = "01"), span(i18n_t("Sample Type"))),
      tagList(span(`data-step` = "02"), span(i18n_t("Sample ID"))),
      tagList(span(`data-step` = "03"), span(i18n_t("Dilution"))),
      tagList(span(`data-step` = "04"), span(i18n_t("Replicates")))
    ),
    choiceValues = c("type", "id", "dilution", "rep"),
    selected = "type"),

div(class = "bs-layer-bulk",
  h6(i18n_t("Bulk action")),
  conditionalPanel("input.active_layer == 'type'",
    selectInput("bulk_type", NULL,
      choices = c("Standard", "Sample", "Blank", "NSB", "B0")),
    actionButton("apply_uniform_type", i18n_t("Set all wells"), class = "btn-default btn-block")
  ),
  conditionalPanel("input.active_layer == 'id'",
    textInput("bulk_id_prefix", NULL, placeholder = "S"),
    actionButton("apply_uniform_id", i18n_t("Set all wells"), class = "btn-default btn-block")
  ),
  conditionalPanel("input.active_layer == 'dilution'",
    # KEEP existing numericInput + apply_uniform_dilution observer
    numericInput("uniform_dilution", NULL, value = 1, min = 0, max = 1, step = 0.01),
    actionButton("apply_uniform_dilution", i18n_t("Set all wells"), class = "btn-default btn-
block")
  ),
  conditionalPanel("input.active_layer == 'rep'",
    textInput("bulk_rep_group", NULL, placeholder = "A"),
    actionButton("apply_uniform_rep", i18n_t("Set all wells"), class = "btn-default btn-block")
  )
)
),

# -- RIGHT: active matrix viewport --
div(class = "bs-matrix-viewport",
  conditionalPanel("input.active_layer == 'type'",
    div(id = "matrix_type_section",
      h4(i18n_t("Sample type")),
      rHandsontableOutput("matrix_type"),
      div(class = "bs-legend",
        span(class="chip", span(class="dot", style="background:rgba(0,158,115,0.4)"),
i18n_t("Standard")),
        span(class="chip", span(class="dot", style="background:rgba(0,114,178,0.4)"),
i18n_t("Sample")),
        span(class="chip", span(class="dot", style="background:#f1f3f5"),
i18n_t("Blank")),
        span(class="chip", span(class="dot", style="background:rgba(204,121,167,0.4)"), "NSB"),
        span(class="chip", span(class="dot", style="background:rgba(230,159,0,0.4)"), "B0")
      )
    )
  ),
  conditionalPanel("input.active_layer == 'id'",

```

```

        div(id = "matrix_id_section",
            h4(i18n_t("Sample ID")),
            rHandsontableOutput("matrix_id")
        )
    ),
    conditionalPanel("input.active_layer == 'dilution'",
        div(id = "matrix_dilution_section",
            h4(i18n_t("Dilution")),
            rHandsontableOutput("matrix_dilution"),
            div(class = "bs-pill is-info", style = "margin-top:10px;",
                i18n_t("Enter the fraction of original sample strength remaining (1 = undiluted)."))
            )
    ),
    conditionalPanel("input.active_layer == 'rep'",
        div(id = "matrix_replicate_section",
            h4(i18n_t("Replicates")),
            rHandsontableOutput("matrix_replicate")
        )
    )
)
)
)
)

```

Important: the `data-step` span trick in `choiceNames` produces the `01 / 02 / 03 / 04` numeric prefix in the layer rail. The CSS in § 1 reads it via `attr(data-step)`.

Do not delete the original 2x2 grid CSS classes (`.matrix-pairs` , `.matrix-pair`) until you have confirmed nothing else references them.

4.2 Sample-type cell coloring (CB-safe)

In the renderer for `matrix_type` (existing `rhandsontable` code), keep the existing `renderer` that paints cells, but swap the color logic to add CSS classes — `cell-standard` , `cell-sample` , `cell-blank` , `cell-nsb` , `cell-b0` — instead of inline colors. The CSS in § 1 paints them.

```

# In your existing rhandsontable() call for matrix_type:
hot_cols(renderer = "function(instance, td, row, col, prop, value, cellProperties) {
  Handsontable.renderers.TextRenderer.apply(this, arguments);
  td.classList.remove('cell-standard', 'cell-sample', 'cell-blank', 'cell-nsb', 'cell-b0');
  if (value === 'Standard') td.classList.add('cell-standard');
  else if (value === 'Sample') td.classList.add('cell-sample');
  else if (value === 'Blank') td.classList.add('cell-blank');
  else if (value === 'NSB') td.classList.add('cell-nsb');
  else if (value === 'B0') td.classList.add('cell-b0');
}")

```

The `green = standard` look in your screenshot must go — green now means PASS only.

§ 5 · Tab 3 — Upload & Preview (with two-column layout + import summary)

Find the Tab 3 block. **Replace** with:

```

fluidRow(

  # -- LEFT (8 cols): file picker + preview + heatmap --
  column(8,
    div(class = "bs-card",
      h3(i18n_t("Upload plate data")),
      p(class = "bs-card-sub", i18n_t("Upload your raw absorbance / counts file. We'll validate and
preview before you continue."))),

      radioButtons("import_method", NULL,
        choices = c("Classic import" = "classic",
          "Visual plate selector" = "visual"),
        selected = "classic", inline = TRUE),

      fileInput("plate_file", NULL,
        accept = c(".xlsx", ".csv", ".txt"),
        buttonLabel = i18n_t("Choose file"),
        placeholder = i18n_t("No file selected")),

      uiOutput("upload_status_ui") # renders the green "Upload complete" pill
    ),

    div(class = "bs-card",
      h4(i18n_t("Plate preview & heatmap")),
      fluidRow(
        column(6,
          h5(i18n_t("Raw values")),
          tableOutput("plate_preview_table") # the existing 8x12 table
        ),
        column(6,
          h5(i18n_t("Heatmap")),
          plotlyOutput("plate_heatmap", height = "320px")
        )
      )
    ),

  ),

  # -- RIGHT (4 cols): import summary --
  column(4,
    div(class = "bs-card",
      h4(i18n_t("Import summary")),
      tableOutput("import_summary_table") # render as bs-report key/value table - see below
    ),
    div(class = "bs-card",
      h4(i18n_t("Validation")),
      uiOutput("import_validation_ui") # uses .bs-pill is-pass/warn/fail
    )
  )
)

```

5.1 Import summary table renderer

```
output$import_summary_table <- renderUI({
  req(shared$import_meta) # your existing reactive – do not change
  m <- shared$import_meta
  tags$table(class = "bs-report",
    tags$tbody(
      tags$tr(tags$td("Format"),      tags$td(class = "num", m$format)),
      tags$tr(tags$td("Wells"),      tags$td(class = "num", paste0(m$wells_filled, " / ",
m$wells_total))),
      tags$tr(tags$td("Partial"),      tags$td(class = "num", if (m$partial) "Yes" else "No")),
      tags$tr(tags$td("Min / Max"),      tags$td(class = "num", sprintf("%.3f / %.3f", m$min, m$max))),
      tags$tr(tags$td("File size"),      tags$td(class = "num", m$size_human))
    )
  )
})
```

5.2 Heatmap palette (Okabe-Ito-derived)

In `reports/plot_functions.R` (or wherever the heatmap is built), find the `RdBu` / `Spectral` / default palette and replace:

```
# OLD: scale_fill_distiller(palette = "RdBu", direction = -1)
# NEW:
scale_fill_gradientn(
  colors = c("#1f3a5f", "#0072b2", "#009e73", "#f0e442"),
  name = "Value"
)
```

If using `plotly::plot_ly(... type = "heatmap")`:

```
colorscale = list(
  c(0.00, "#1f3a5f"),
  c(0.33, "#0072b2"),
  c(0.66, "#009e73"),
  c(1.00, "#f0e442")
)
```

§ 6 • Tabs 4 & 5 (unchanged from v1 spec)

The v1 guide's § 5.2 / § 5.3 stand. The CSS in § 1 of this v2 file already provides the form rows, format tiles, and pre-flight panel components — wire your existing inputs into them by renaming wrapper classes (`bs-card`, `bs-section-row`, `bs-tile-grid`, `bs-tile-check`, `bs-preflight`).

§ 7 • Report (`reports/unified_analysis_template.Rmd`)

Same as v1 § 6. The CSS classes `.bs-kpis` / `.bs-kpi` / `.bs-pill` / `table.bs-report` / `.report-title` are already in `style.css` — your Rmd just needs to emit the right HTML.

§ 8 • Acceptance checklist (must all pass)

Block delivery if any of these fail:

- ☐ Wizard tab **always** legible (not white on white). Inactive label is ink-3, active is navy on accent-soft, text is done is ink-2 with check.
- ☐ Stepper has visible 1-px connectors between consecutive tabs.
- ☐ Top bar renders as one row:

[B] Bioassay Suite v0.9.0 Guided tour · Docs | English ▾

.
- ☐ Quick Start renders as 3 **tiles** (not Bootstrap buttons), each with a colored 4-px left rail.
- ☐ Tab 1 Assay/Standards: every section heading + description sits in a left **next to** its inputs in the right 280-px column.
- ☐ Tab 2 Plate Layout: vertical layer switcher on the left (240 px) with

01 Sample Type / 02 Sample ID / 03 Dilution / 04 Replicates

, the active row painted with

--c-accent-soft

. Right **single, full-width matrix** shows a **matrix** with
- ☐ Sample-type **not bright green/blue** — they use the muted Okabe-Ito tints from

.cell-standard / .cell-sample / .cell-blank / .cell-nsb / .cell-b0

.
- ☐ Tab 3 **two-column layout** with Plate preview & Heatmap **side by side** in the left card, and an Import Summary card on the right.
- ☐ Tab 3 heatmap palette is navy → blue → green → yellow (no red, no Spectral).
- ☐ Switch all four layers in Tab 2 — edits in the inactive matrices are **preserved**.
- ☐ No console errors on

shiny::runApp()

.
- ☐ CVD simulation pass (Chrome DevTools → Rendering → Emulate vision deficiencies → deuteranopia + protanopia): pass/warn/fail pills and curve markers stay distinguishable.

§ 9 • If you get stuck

Stop. Open the visual mockup

Bioassay Suite – GUI refresh.html

 in a browser, focus the artboard you're working on (click its label), screenshot it, and compare to your build. If the gap is structural (markup-level) rather than stylistic (CSS-only), **the markup is the redesign** — change it.

If a Shiny / Bootstrap 3 limitation makes the markup change look impossible, message the designer instead of falling back to the old structure with new colors. That fallback is what produced v1.